Vehicular Communications 49 (2024) 100801

Contents lists available at ScienceDirect

Vehicular Communications

journal homepage: www.elsevier.com/locate/vehcom

Discharging driven energy sharing protocol for V2V communication in vehicular energy networks

Qinghua Tang [a], Demin Li [a,*], Yihong Zhang [a,*], Xuemin Chen [b]

[a] College of Information Science and Technology, Engineering Research Center of Digitized Textile and Apparel Technology, Ministry of Education, Donghua University, Shanghai 201620, China

[b] Department of Engineering, Texas Southern University, Houston, TX 77004, USA

ARTICLE INFO

Keywords:
Electric vehicles (EVs)
Vehicular energy network (VEN)
Energy sharing
Vehicular ad-hoc networks (VANETs)
Discharging driven

ABSTRACT

With the growing concern over range anxiety among electric vehicle (EV) owners due to limited battery capacity and sparse charging infrastructure, EV-to-EV (V2V) energy sharing emerges as a crucial solution to extend driving range. Leveraging the vehicular energy network (VEN) enabled by dynamic wireless power transfer (DWPT) technology, energy sharing among EVs in motion becomes feasible. However, the effective establishment of communication and identification of suitable V2V energy sharing pairs pose significant challenges, particularly for EVs with discharging demands. To address this challenge, we propose a new discharging driven energy sharing protocol based on vehicular ad-hoc networks (VANETs). Firstly, we present a routing approach for transmitting discharging information through VANETs, considering key factors such as distance, state of charge, number of neighbor vehicles, and vehicle speed. This routing scheme facilitates efficient relay node selection on road segments and intersections, ensuring optimal communication paths. Subsequently, we formulate a charging requester selection model to identify the most suitable requester for energy sharing. This model optimizes individual utility while accounting for the state of charge of the requesters, ensuring a comprehensive and inclusive approach to V2V energy sharing. Finally, we develop an acknowledgment message transmission scheme to ensure the completion of selection acknowledgment between discharging EVs and charging EVs. This scheme includes provisions for recovery in case of forwarding link failures, ensuring robust communication in dynamic vehicular environments. Extensive simulations conducted using network simulator 2 (NS-2) demonstrate the superior performance of the proposed protocol in terms of packet delivery ratio, end-to-end delay, and overall V2V energy sharing efficiency.

1. Introduction

The vehicle energy network (VEN) leverages the existing road infrastructure to facilitate efficient energy transmission through electric vehicles (EVs) across vast geographical areas [1]. Comprising a fleet of mobile EVs and roadside energy nodes equipped with dynamic wireless power transfer (DWPT) facilities [2], VEN relies on DWPT technology to transfer power wirelessly to EVs while they are in motion [3], [4]. This capability allows EVs with surplus battery energy to share it with neighboring vehicles via DWPT, thereby enabling wireless energy transmission within VEN [5]. VEN introduces the concept of EV-to-EV (V2V) energy scheduling while driving, offering a novel approach to EV charging. In addition to drawing energy from the grid, EVs can function as mobile energy sources, sharing excess charge with other vehicles that require it. This approach reduces dependency on centralized charging stations, enhances energy availability, and alleviates congestion at charging points. V2V communication facilitates the exchange of energy-related information between EVs while they are in motion. By enabling real-time communication, V2V technology ensures efficient energy sharing as vehicles navigate through varying traffic conditions and energy demands. For instance, a discharging EV (DEV) can transmit discharging information to nearby vehicles, prompting them to request a charge as needed. Thus, establishing V2V communication during vehicle motion is essential for effective coordination of V2V energy scheduling.

Vehicular ad-hoc networks (VANETs) are known for their self-organizing nature and scalability, enabling the dissemination of information through wireless multi-hop communication between vehicles-to-

* Corresponding authors.
E-mail addresses: tangqinghua@mail.dhu.edu.cn (Q. Tang), deminli@dhu.edu.cn (D. Li), zhangyh@dhu.edu.cn (Y. Zhang), xuemin.chen@tsu.edu (X. Chen).

https://doi.org/10.1016/j.vehcom.2024.100801
Received 11 July 2023; Received in revised form 31 March 2024; Accepted 26 May 2024
Available online 4 June 2024
2214-2096/© 2024 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

vehicles (V2V) and vehicles-to-infrastructure (V2I) [6], [7], [8]. In the context of VEN, VANETs play a crucial role in facilitating energy sharing among EVs [9], serving as the communication backbone for transmitting energy-related information and acknowledging V2V energy sharing pairs. Researchers have explored various VANET-based solutions for enhancing EV charging and energy sharing. For instance, Bautista et al. utilized a multi-metric map-aware routing protocol proposed in [10] for VANETs to enable multi-hop transmission of service messages between EVs and roadside units (RSUs) to secure anticipated charging slot reservations [11]. Similarly, Li et al. proposed a V2V charging navigation system leveraging flexible VANET communication modes to ensure reliable transmission of charging information between navigation control centers, distributed stopping locations, and moving EVs [12]. Once V2V energy information communication is established using VANETs, the selection of suitable charging pairs becomes a critical issue in realizing V2V energy sharing. Existing works have primarily focused on minimizing the costs associated with DEVs and charging EVs (CEVs) to optimize the scheduling of DEVs and CEVs [13], [14]. However, to the best of our knowledge, these works have not thoroughly addressed the selection of next-hop relay nodes between intersections and at intersections to ensure reliable delivery of discharging information via multi-hop V2V communication. Moreover, existing approaches often overlook the state of charge (SoC) of the CEV when determining V2V energy sharing pairs, which may lead to unrealistic scenarios. The contributions of this paper are as follows:

- We propose a new discharging information transmission routing mechanism leveraging VANETs to identify CEVs. This routing protocol operates along road segments and intersections, selecting the next-hop relay node based on a comprehensive metric value. This metric incorporates factors such as distance, SoC, the number of neighboring vehicles, and speed, enabling efficient CEV discovery and communication.
- We develop an optimization model aimed at selecting the most suitable CEV for V2V energy sharing in VEN. This model takes into account both individual utility and the remaining SoC of the charging requester as primary objectives. Additionally, we introduce a selection CEV algorithm derived from this optimization model.
- We examine the communication process between DEV and CEV pairs and propose a request and acknowledgment message transmission scheme. This scheme is designed to address potential challenges such as forwarding link failures, ensuring the successful completion of selection and acknowledgment between DEV and CEV pairs.

The remaining sections are organized as follows: Section 2 discusses state-of-the-art work on V2V energy trading problem. The system model is given in Section 3. Section 4 proposes the discharging driven energy sharing protocol design. Performance evaluations are shown in Section 5. Section 6 presents the conclusion and future work.

2. Related work

Existing works on V2V energy sharing address various challenges associated with large-scale EV charging, aiming to alleviate charging pressure during peak demand periods [15],[16]. These works mainly tackle V2V charging service problems through centralized servers or distributed approaches based on V2V communication, aiming to realize more intelligent charging and discharging behaviors.

Abdolmaleki et al. introduced vehicle-to-vehicle wireless power transfer between the EVs [17]. Power transfer between EVs is based on the same travel path of the supplier and the receiver. A dynamic programming algorithm was proposed to solve vehicle routing, scheduling, and matching problems on V2V wireless power transfer (WPT) platforms to enable as many trips as possible while minimizing total energy costs for all system users. Zhang et al. introduced cooperative V2V charging between EVs that demand power act and EVs with extra power act [18]. Using the control center to collect EVs information from energy consumers and energy providers, they made smart charging/discharging decisions for involved EVs. According to the stable matching concept based on the individual rationality of each involved EV, two stable V2V matching algorithms were proposed to obtain EV-consumer-optimal and EV-provider-optimal stable V2V matching respectively. Bulut et al. proposed a central mechanism for V2V charging pair allocation [19]. EVs with surplus energy can be used when stationary, and EV that needs energy submits a charging request to the central server. The authors proposed two matching methods based on system benefit and user preference to solve the matching challenge of demander EVs and the charge suppliers. Shurrab et al. introduced a comprehensive energy management framework for V2V energy sharing [20], [21]. Their framework encompasses the entire process from the publication of a charging request to the final matching and actual energy transfer between EVs. The authors established preferences for EVs by evaluating criteria such as charging cost, charging time, energy demand fulfillment rate, and user satisfaction. Based on these preferences and user satisfaction levels, they determine V2V charging pairs. This approach aims to optimize the allocation of charging resources and enhance user satisfaction in the V2V energy sharing system. Ko et al. proposed a software-defined EV-to-EV charging framework (SD-E2EC) with mobile aggregators, where a charging controller determines the charging monetary cost and location to CEVs and location, and the monetary reward for DEVs [22]. Then, CEVs and DEVs decide whether to accept the charging controller's advice to determine V2V charging pairs.

Said et al. proposed a scheme for charging and discharging services for EVs [23]. RSUs broadcast EV charging and discharging services over the control channel and initiate negotiation during the service channel. They designed corresponding algorithms to manage the booking and negotiation of EV charging and discharging services. Kosmanos et al. focused on addressing anxiety regarding the driving range of EVs. They proposed a dynamic wireless charging method for EVs, utilizing buses or trucks as energy sources [24]. In their approach, EVs follow part of the mobile energy disseminator (MED) routes for dynamic wireless charging, aiming to extend the travel range of EVs. They studied the intelligent routing problem of EVs needing charging based on V2V communication, which improved travel time and driving range. Abualola et al. proposed a V2V charging allocation protocol [25]. They employed Quality of Service-Optimized Link State Routing (QoS-OLSR) to discover routing paths for disseminating providers' announcements, allowing consumers in the network to discover nearby providers. Upon receiving the provider announcement, the consumer initiates the charging pair allocation process by computing the payment value and transmitting an OFFER message to the chosen provider [25]. Subsequently, the provider begins the charging pair allocation process by evaluating the payoff value based on the received OFFER message and dispatches an ACK message to the selected consumers, thereby finalizing the allocation of V2V charging pairs. Yucel et al. introduced a method for privately-preserving distributed stable matching of suppliers [26]. Using partial homomorphic encryption, neither the demander nor the supplier has access to the actual location of the other party; the demander only needs to know the distance to the supplier and creates a preference list based on ascending order of distance. Suppliers are then assigned according to this preference list. Chatterjee et al. ranked vehicles based on five criteria: cost per unit charge, the trust value, the amount of available charge, estimated travel time to the requesting vehicle, and charge transfer rate [27]. They assigned different weights to these criteria and proposed a V2V Distributed Charge Sharing (V2VDisCS) approach to optimize matching cardinality and minimize matching cost. In V2VDisCS, acceptors rank reachable donors and send REGISTER messages containing ranked lists. Donors select acceptors based on priority and send CONFIRM messages. Acceptors then select the lowest-ranked donor and send MATCHED messages within a specified time frame. We summarize the related work and our work in Table 1.

Overall, research on V2V energy sharing among EVs is advancing, gradually alleviating the range anxiety of EVs. In regions with limited charging infrastructure or during peak charging hours, EVs with surplus power establish V2V energy sharing pairs through multi-hop communi-

Table 1
Characteristics in previous works.

Features / Works	System architecture	Methodology	Contributions
Ref. [17]	Centralized architecture	Dynamic programming algorithm for V2V wireless power transfer routing, scheduling, and matching	Optimization of energy costs while maximizing trips for all users
Ref. [18]	Centralized architecture	Cooperative V2V charging between EVs, smart charging/discharging decisions based on centralized control center	Stable matching algorithm based on individual rationality for optimal V2V pairing
Ref. [19]	Centralized architecture	Central mechanism for V2V charging pair allocation, charging requests to central server	Matching methods based on system benefit and user preference
Ref. [20], [21]	Centralized architecture	Comprehensive energy management framework for V2V energy sharing, preference-based EV pairing	Optimization of charging resources allocation and user satisfaction
Ref. [22]	Centralized architecture	Software-defined EV-to-EV charging framework with mobile aggregators, charging controller advice for V2V charging pairs	Decision-making for charging monetary cost and location
Ref. [23]	Distributed architecture	RSU-broadcasted EV charging and discharging service scheme, negotiation algorithms for service booking	Management of EV charging and discharging service negotiation
Ref. [24]	Distributed architecture	Dynamic wireless charging method utilizing buses or trucks as energy sources, intelligent routing based on V2V communication	Expansion of EV travel range and improvement of travel time
Ref. [25]	Distributed architecture	V2V charging allocation protocol using QoS-OLSR routing, charging pair allocation process based on provider announcements	Discovery of nearby consumers and charging pair allocation based on payment and payoff value
Ref. [26]	Distributed architecture	Privately-preserving distributed stable matching of suppliers using partial homomorphic encryption	Matching of suppliers based on distance preference lists while preserving privacy
Ref. [27]	Distributed architecture	Distributed heuristics approach for V2V Distributed Charge Sharing	Determination of V2V charging pairs to maximize matching cardinality and minimize matching cost
our work	Distributed architecture	V2V energy sharing pair is allocated using the discharge information transmission and selection and acknowledgment scheme.	Discover CEVs and select the CEV based on individual utility and the remaining SoC of the charging requester

cation, offering a solution to mitigate charging pressure. However, most existing works focus on determining V2V energy pairs based on the distance between CEV and DEV, or they evaluate individual rationality and user satisfaction by considering discharging utility and charging costs. In real-world scenarios, the SoC level of CEVs becomes critically important when establishing V2V energy sharing pairs. By considering the SoC level of the CEV, it is possible to prioritize service for EVs with lower charge levels during emergency charging situations, thus enhancing the system's responsiveness to urgent needs. Moreover, the dissemination of discharging information between vehicles and the delivery of selection and acknowledgment messages to enable on-the-move vehicle charging pose significant challenges in VEN, especially in the absence of a reliable communication infrastructure.

3. System model

In this section, we consider the energy sharing system in VEN. Fig. 1 illustrates an example of a road segment, which primarily includes the energy storage devices equipped with wireless charging/discharging tracks, and moving EVs. The descriptions of notations used in the following are shown in Table 2.

Road Network: The road network can be described as a graph $\mathcal{G}(\boldsymbol{V}, \boldsymbol{\Theta})$, where $\boldsymbol{V}$ denotes the set of all the road intersections, $\boldsymbol{\Theta}$ denotes the set of all the road segments, and for $m, n \in \boldsymbol{V}$, if there is an accessible road, we express as (m, n), $(m, n) \in \boldsymbol{\Theta}$. L_{mn} is the length of segment (m, n).

Energy storage devices: In VEN, the energy storage devices are deployed on each road segment, featuring a lane with a charging track and a lane with a discharging track for dynamic energy sharing with EVs through DWPT. These charging/discharging tracks consist of coils buried under the road surface and are connected by a common bus with energy storage devices. Consequently, EVs can be charged or discharged without the need to stop or slow down. Any energy deficit can be compensated for by the energy storage system, while excess energy can be stored for future use.

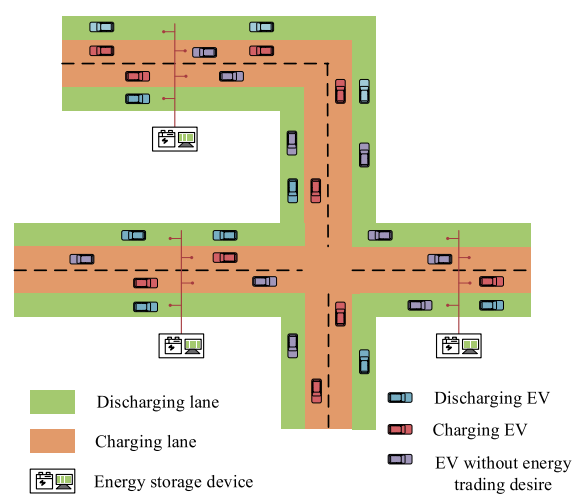


Fig. 1. Schematic of charging and discharging facilities with energy storage at road.

EVs: In the paper, EVs are categorized into three groups: 1) EVs that require power, designated as CEVs and denoted as $EVC_i, i = 1, 2, \ldots, N$; 2) EVs with surplus power, designated as DEVs and denoted as $EVP_j, j = 1, 2, \ldots, M$; 3) EVs uninterested in participating in current energy trading activities. Within the VEN, EVs with excess or insufficient battery energy can engage in energy exchange using power tracks installed in energy storage devices via DWPT. Therefore, DEVs can identify suitable CEVs for energy sharing within a specific lane.

For better analysis, we make the following basic assumptions:

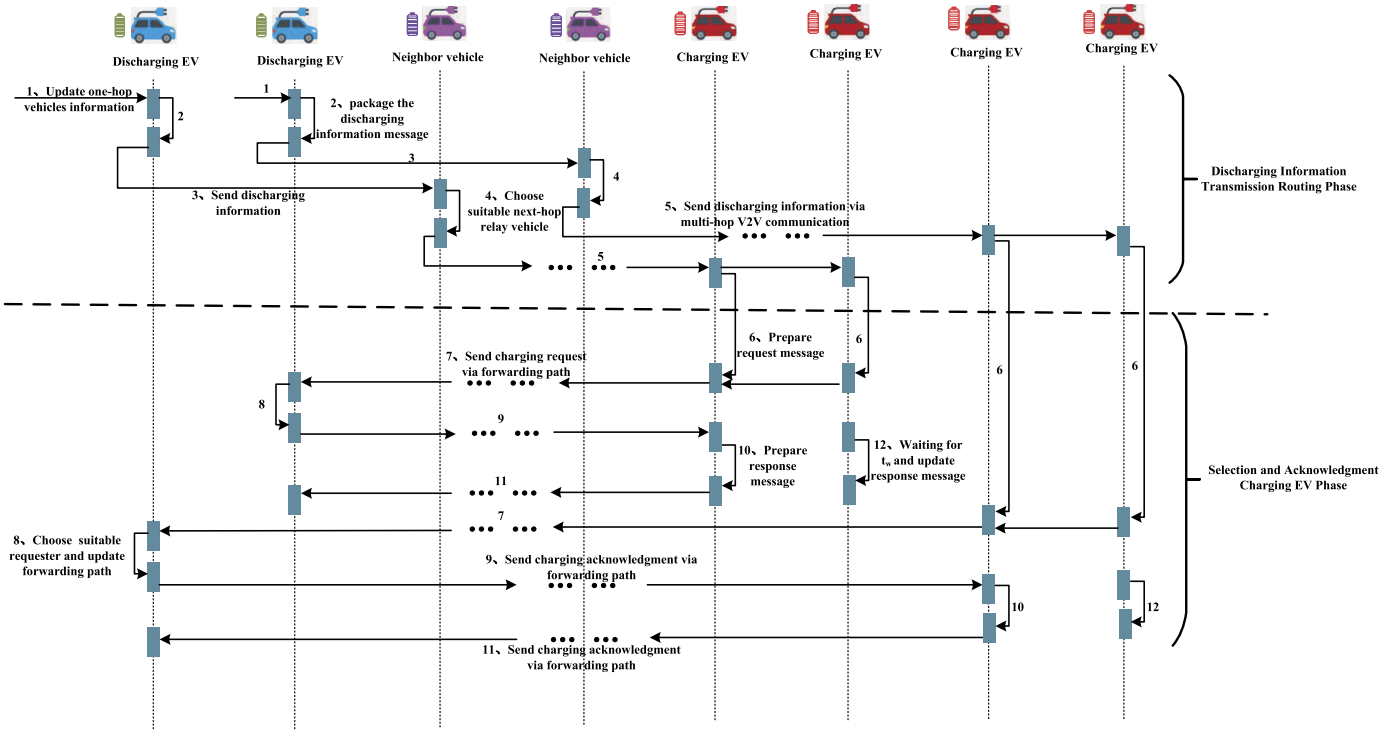


Fig. 2. Illustration of discharging driven energy sharing protocol procedure.

Table 2
The main mathematical notations.

Notation	Description
$\boldsymbol{V}$	Road node set in graph $\mathcal{G}$.
$\boldsymbol{\Theta}$	Road segment set in graph $\mathcal{G}$.
(m,n)	Charging/discharging service segment.
P^j	The travel path of the EVP_j.
p^j	The discharging path of the EVP_j.
p^i	The charging path of the EVC_i.
R_j	EVP_j available charging range.
r	Vehicle wireless communication range.
t_{hop}	Average wireless transmission delay per hop.
$d_{A,B}$	The distance between vehicle A and vehicle B.
$N_{B\|A}$	The number of neighbors common to the vehicle A and the vehicle B.
N_B	The number of neighbor vehicles of the vehicle B.
v_A	The speed of vehicle A.
$T(\cdot)$	The function calculate travel time.
q_{mn}	The number of current vehicles on the road segment (m,n).
q_{mn}^{max}	The number of maximum vehicles on the road segment (m,n).
$Dis(\cdot)$	The function calculate travel distance.
P_i^j	The discharging utility of the EVP_j.
b_j	The power amount that can be sold by the EVP_j.
a_i	The power amount requested by the EVC_i.

• Each EV has a unique ID and is equipped with an on-board unit (OBU) and a global positioning system (GPS) map for V2V communication and obtaining location information.

• Each EV obtains information about neighboring vehicles, including the number of neighbors, location, speed, and direction, through wireless communication in VANETs by periodically sending Hello packets.

• The number of vehicles on a road segment can be obtained via a induction loop beside intersections.

• The communication range of vehicles, denoted as r, is greater than or equal to the width of the road segment. Currently, the communication range of vehicles is typically set at 250 meters [10], [11], [12], [25], which exceeds the width of roads in most urban areas.

• Some EVs are equipped with bidirectional chargers and DWPT devices for wireless charging and discharging. Our work focuses on determining V2V energy sharing pairs among EVs equipped with both bidirectional chargers and DWPT technology, which have both charging and discharging requirements.

4. Discharging driven energy sharing protocol design

4.1. Overview

A discharging driven energy sharing protocol is proposed to discover nearby CEVs and facilitate V2V energy sharing. Fig. 2 illustrates the protocol procedure, which includes the discharging information transmission routing phase and the selection and acknowledgment CEV phase. Initially, the DEV transmits discharging information to vehicles along the travel path using multi-hop V2V communication, as depicted in procedures 1-5. Upon receiving the discharging information message, the CEV sends a charging request message to the DEV, as illustrated in procedures 6-7. Subsequently, the DEV selects the optimal requester by calculating the V2V energy benefit and responds to the most suitable requester with a charging acknowledgment message, as shown in procedures 8-9. Additionally, the selected CEV acknowledges the chosen DEV, which replies with a charging acknowledgment message, as demonstrated in procedures 10-11. The unselected CEV updates the response message to select the other DEV, as depicted in procedure 12. Since CEVs and DEVs may not always be within each other's communication range, one-hop V2V communication may not suffice to complete the selection and confirmation of V2V energy sharing. Hence, the utilization of multi-hop and carry-and-forward communication mechanisms becomes essential. These mechanisms facilitate communication between distant vehicles and ensure the successful completion of the energy sharing process. In this section, we discuss the establishment of communication between CEVs and DEVs using multi-hop and carry-and-forward communication to facilitate the selection and confirmation process. Finally, the DEV and selected CEV can engage in energy sharing via DWPT with the assistance of power tracks installed on energy storage devices.

4.2. Discharging information transmission routing phase

The discharging information transmission routing phase is crucial for finding a suitable CEV. Here, we use R_j to denote the distance between the DEV and the travel destination. The sending range of discharging information is defined as the area within a circle of radius R_j.

The discharging information includes DEV location, timestamp, energy unit selling price, available power, speed, and path. DEV set the activity range of the discharging information message packet to R_j based on the distance to the travel destination. Therefore, the relay vehicle is selected from the one-hop neighbor vehicles of each packet carrier vehicle within R_j. Only the vehicle selected as the relay node forwards the discharging information message, while the other one-hop neighbor vehicles only accept and do not forward.

In this section, we further optimize the discharging information transmission routing strategy, which aims to discover CEVs. Abualola et al. used modified QoS-OLSR [25] to send provider information and set the time to live of provider information messages to end the delivery. However, they did not discuss in detail the transmission of discharging information to determine V2V energy sharing. Additionally, When the relay vehicle is selected, the forwarding willingness of the neighbor vehicle is not considered, which may interrupt the information forwarding. As a result, we discuss in detail the discharging information transmission in the road network.

4.2.1. Relay node selection between intersections

In this section, we discuss how to select suitable next-hop relay vehicles between intersections to deliver discharging information packets. The packet carrier vehicles are within the range R_j, and the selection of next-hop relay vehicles between intersections is analyzed as follows:

Case 1: The packet carrier vehicle has multiple one-hop neighbor vehicles.

In this case, given the uncertainty and dynamics of the vehicular environment, we optimize the selection of relay nodes by considering factors such as position, SoC, the number of one-hop neighboring vehicles, and vehicle speed. We define a comprehensive metric S to select the next-hop relay vehicle with the maximum S. The metric S is composed of the distance ratio, the SoC ratio, the ratio of neighboring vehicles, and the ratio of the speed difference, all of which are crucial for effective discharging information transmission. Thus, the expression for the metric S expression is as follows:

$$S = \alpha_1 \frac{d_{A,B}}{r} + \alpha_2 \frac{SoC_B}{SoC_B^{max}} + \alpha_3 \frac{N_B - N_{B|A}}{N_B} + \alpha_4 (1 - \frac{|v_B - v_A|}{v_A}), \quad (1)$$

here, α_1, α_2, α_3 and α_4 are weight factors, $\alpha_1, \alpha_2, \alpha_3, \alpha_4 \in [0,1]$ and $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 1$. $d_{A,B}$ is the distance between vehicle A and vehicle B. Vehicle A is the packet carrier vehicle, and vehicle B is its one-hop neighbor. Therefore, in Eq. (1), $0 < d_{A,B} < r$. $\frac{d_{A,B}}{r}$ reflects the distance relationship between adjacent vehicles, and to reduce the number of hops, the vehicle B with a large $\frac{d_{A,B}}{r}$ value is better. $\frac{SoC_B}{SoC_B^{max}}$ is the SoC ratio of vehicle B, which is used to evaluate the forwarding willingness level of vehicle B. $N_{B|A}$ is the number of neighbors common to the vehicle A and the vehicle B. N_B is the number of neighbor vehicles of the vehicle B. $\frac{N_B - N_{B|A}}{N_B}$ reflects the ratio of neighbors of vehicle B that are not also neighbors of vehicle A. v_A and v_B are the speed of vehicle A and B. $\frac{|v_B - v_A|}{v_A}$ reflects the speed difference between vehicle A and neighbor vehicle B. To enhance the stability of the vehicle's communication link, it's beneficial to have a relatively large number of neighboring nodes and SoC values for the vehicle, along with a relatively smaller speed difference between vehicles. Therefore, selecting relay vehicle with relatively large $\frac{N_B - N_{B|A}}{N_B}$, $\frac{SoC_B}{SoC_B^{max}}$, and $1 - \frac{|v_B - v_A|}{v_A}$ values is helpful to improve link stability. The calculation of the S value for selecting the next-hop relay node is considered only when the velocity of vehicle A is non-zero and the number of one-hop neighbor nodes for vehicle B is non-zero. Thus, in Eq. (1), $v_A > 0$ and $N_B > 0$.

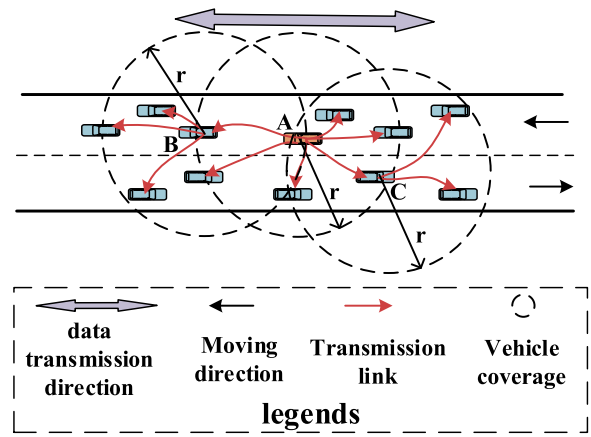


Fig. 3. DEV A has multiple one-hop neighbor vehicles.

We assume that the coordinates of vehicles A, B at time t are (x_A, y_A) and (x_B, y_B). The distance between vehicles can be approximated as,

$$d_{A,B} = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2}. \quad (2)$$

Case 1.1: The packet carrier vehicle is DEV.

As shown in Fig. 3, the multiple one-hop vehicles include both moving vehicles in the same direction as the direction of the discharging information transmission and moving vehicles in the opposite direction to the direction of the discharging information transmission within the communication range of DEV A. Since discharging information needs to be transmitted along each moving direction of the road around the DEV, one relay vehicle B needs to be selected from the neighbor vehicles moving in the same direction, and one relay vehicle C needs to be selected from the neighbor vehicles moving in the opposite direction of the DEV according to the S value.

Case 1.2: The packet carrier vehicle is relay vehicle.

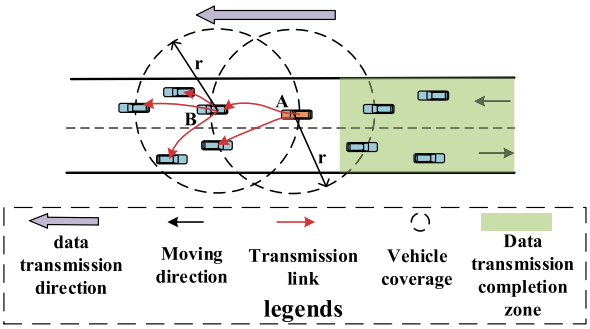
In this case, we need to consider the relationship between the moving direction of two-hop neighbor vehicle and the discharging information transmission direction to select the next-hop relay vehicle.

Case 1.2(a): The multiple one-hop vehicles include moving vehicles in the same direction as the direction of the discharging information transmission.

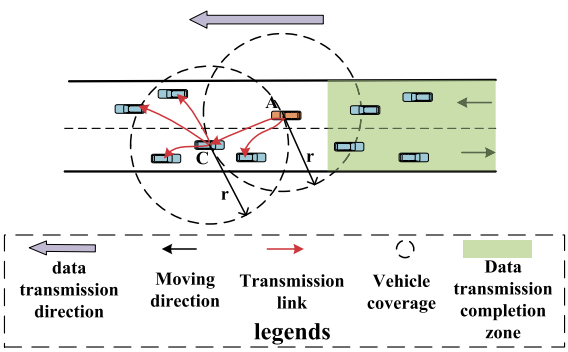
As depicted in Fig. 4(a), there are one-hop vehicles moving in the same direction as the direction of the discharging information transmission within the communication range of relay vehicle A. The discharging information is transmitted along the road segment, so the next-hop relay vehicle B is selected with the maximum S value among the moving neighbor vehicles in the same direction as the discharging information transmission.

Case 1.2(b): The multiple one-hop vehicles only include moving vehicles in the opposite direction to the discharging information transmission.

As depicted in Fig. 4(b), there are multiple one-hop vehicles moving in the opposite direction to the discharging information transmission within the communication range of relay vehicle A. We assess the presence of a next-hop vehicle and whether its moving direction aligns with the discharging information transmission direction. If such a vehicle exists, the next step involves selecting the next-hop relay vehicle based on the S metric. Otherwise, relay vehicle A solely transmits the discharging information to one-hop vehicle and employs the carry-and-forward mechanism until an appropriate relay vehicle is identified within its communication range.



(a) The multiple one-hop vehicles include moving vehicles in the same direction as the direction of the discharging information transmission.



(b) The multiple one-hop vehicles only include moving vehicles in the opposite direction to the discharging information transmission.

Fig. 4. Relay vehicle A has multiple one-hop neighbor vehicles.

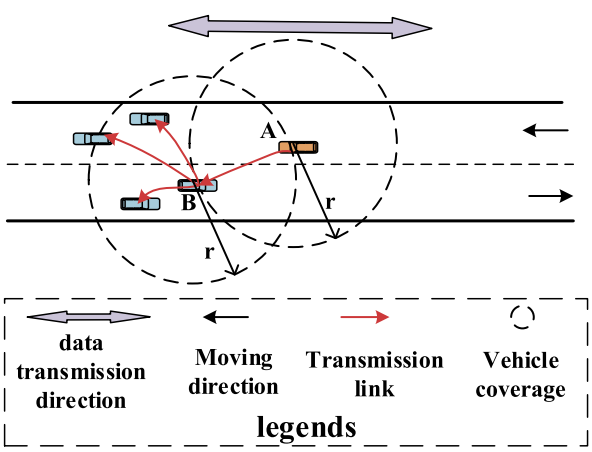


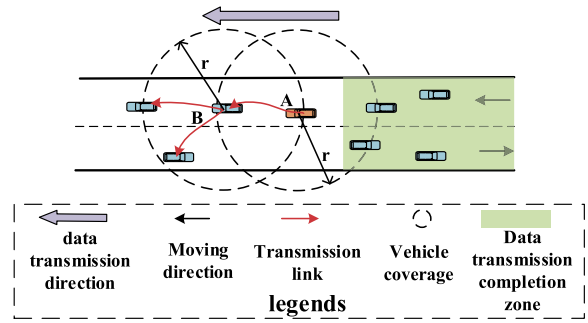
Fig. 5. DEV A only has one one-hop neighbor vehicle.

Case 2: The packet carrier vehicle only has one one-hop neighbor vehicles.

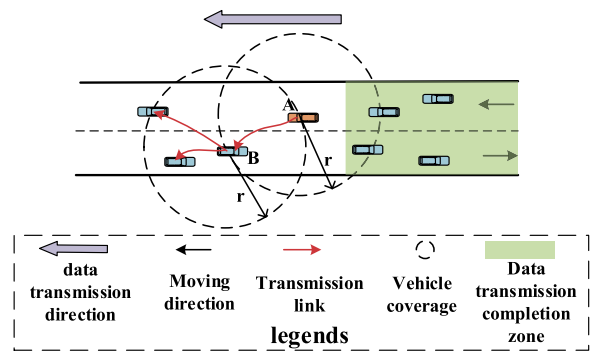
In this case, we must consider whether to designate the one-hop vehicle as the next-hop relay vehicle.

Case 2.1: The packet carrier vehicle is DEV.

As shown in Fig. 5, there is only one one-hop neighbor vehicle B within the communication range of DEV A. First, we determine whether the condition $\frac{SoC_B}{SoC_B^{max}} \geq \frac{SoC_A}{SoC_A^{max}}$ is satisfied. If this condition is met, the next consideration is whether there is a next-hop vehicle within the communication range of vehicle B, and whether its movement aligns with the direction of discharging information transmission. If these conditions are fulfilled, vehicle B is chosen as the next-hop relay vehicle. Otherwise, DEV A will solely transmit the discharging information to vehicle B and resort to the carry-and-forward mechanism until an appropriate relay vehicle is located within its communication range.



(a) The vehicle's direction of movement aligns with the direction of discharging information transmission.



(b) The vehicle's direction of movement is opposite to the direction of discharging information transmission.

Fig. 6. Relay vehicle A only has one one-hop neighbor vehicle.

Case 2.2: The packet carrier vehicle is relay vehicle.

In this case, it's necessary to consider whether to designate the one-hop vehicle as the next-hop relay vehicle by evaluating the movement direction of the two-hop neighbor vehicles.

Case 2.2(a): The vehicle's direction of movement aligns with the direction of discharging information transmission.

As shown in Fig. 6(a), there is only one one-hop vehicle B within the communication range of relay vehicle A. We determine the $\frac{SoC_B}{SoC_B^{max}}$ of vehicle B. If $\frac{SoC_B}{SoC_B^{max}} \geq \frac{SoC_A}{SoC_A^{max}}$, vehicle B is selected as the next-hop relay vehicle. Otherwise, relay vehicle A will solely transmit the discharging information to vehicle B and employ the carry-and-forward mechanism until an appropriate relay vehicle is located within its communication range.

Case 2.2(b): The vehicle's direction of movement is opposite to the direction of discharging information transmission.

As shown in Fig. 6(b), there is only one one-hop vehicle B within the communication range of relay vehicle A. First, we determine the $\frac{SoC_B}{SoC_B^{max}}$ of vehicle B. If $\frac{SoC_B}{SoC_B^{max}} \geq \frac{SoC_A}{SoC_A^{max}}$, the next step is to consider whether there is one-hop vehicle and the moving direction is the same as the discharging information transmission direction within the range of vehicle B. If vehicle B meets the aforementioned criteria, it is selected as the next-hop relay vehicle. If vehicle B does not meet these conditions, relay vehicle A only sends discharging information to vehicle B and adopts the carry-and-forward mechanism until a suitable relay vehicle is found within its communication range.

Case 3: The packet carrier vehicle has no one-hop vehicles in the communication range.

In this case, the packet carrier vehicle adopts the carry-and-forward mechanism until a suitable relay vehicle is found within its communication range.

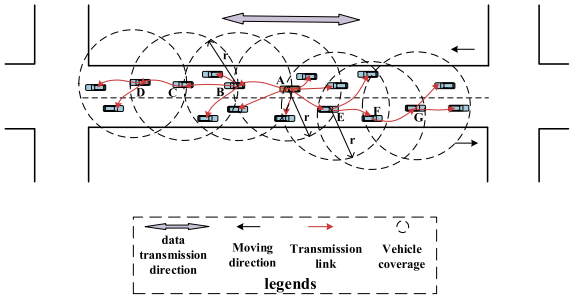


Fig. 7. Illustration of the discharging information is transmitted at road segment.

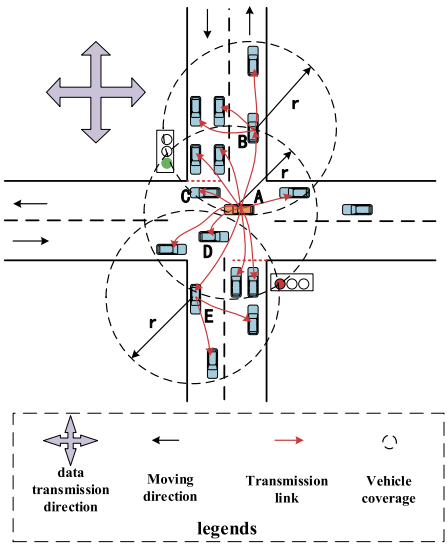


Fig. 8. DEV A deliver packets at intersection.

Based on the above analysis of discharging information transmission between intersections, the DEV sends the discharging information message to the one-hop neighbor vehicle. Then, the discharging information message is forwarded along both sides of the moving direction according to the next-hop relay vehicle selection method mentioned above. As shown in Fig. 7, the discharging information message is forwarded along routing path $A \rightarrow B \rightarrow C \rightarrow D$ and $A \rightarrow E \rightarrow F \rightarrow G$.

4.2.2. Relay node selection at the intersection

In this section, we discuss how to select suitable next-hop relay vehicles to deliver discharging information at intersections. The packet carrier vehicles are in the range R_j, and the selection of relay vehicles at intersections is analyzed as follows:

Case 4: The packet carrier vehicle is DEV.

As shown in Fig. 8, at intersections, when the packet carrier vehicle is DEV, each direction of the road needs to select the next-hop relay vehicle with a maximum S value to deliver the discharging information data packet (i.e., vehicle B, C, D, E).

Case 5: The packet carrier vehicle is relay vehicle.

As shown in Fig. 9, at intersections, when the packet carrier vehicle is a relay vehicle, each direction of the road, except the one covered by discharging information, needs to select the next-hop relay vehicle with the maximum S value to deliver the discharging information data packet (i.e., vehicle B, C, D).

4.2.3. Routing algorithm

Based on the analysis in Section 4.2.1 and Section 4.2.2, we propose a discharging information transmission routing algorithm, which

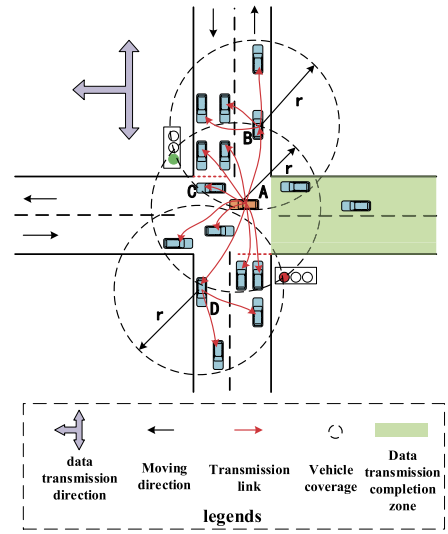


Fig. 9. Relay vehicle A deliver packets at intersection.

includes the transmission of discharging information both between intersections and at intersections. The algorithm also records information about the relay vehicle node, as shown in Algorithm 1.

Each vehicle maintains a neighbor table, which contains the SoC, position, speed, moving direction, and number of neighbor vehicles. This neighbor table is updated by exchanging Hello messages periodically [28]. In Algorithm 1, CA represents the correlation between the vehicle's direction of movement and the direction of discharging information transmission, while DC represents the number of one-hop vehicles of the packet carrier vehicle. $CA = 11$ indicates that the vehicle's direction of movement aligns with the direction of discharging information transmission. $CA = 10$ means that there are both moving vehicles in the same direction with the direction of the discharging information transmission and moving vehicles in the opposite direction as the direction of the discharging information transmission. $CA = 00$ indicates that the vehicle's direction of movement is opposite to the direction of discharging information transmission. $DC = 11$ indicates there are multiple one-hop vehicles of the packet carrier vehicle. $DC = 10$ indicates there is one one-hop vehicle of the packet carrier vehicle. $DC = 00$ indicates there is no one-hop vehicle of the packet carrier vehicle. Additionally, CZ indicates whether vehicles received discharging information from the DEV. Specifically, $CZ = 11$ indicates that vehicles received discharging information from the DEV, whereas $CZ = 00$ indicates that they have not received the discharging information. Lines 9 to 21 describe the process of selecting the next-hop relay node for DEV. Lines 22 to 43 describe the process of selecting the next-hop relay node for the relay vehicle. Lines 44 to 46 describe the process of recording the information of the relay vehicle node.

4.3. Selection and acknowledgment CEV routing phase

When the CEV receives a discharging information message, it sends a charging request message to the DEV. If the CEV receives discharging information from multiple DEVs, it will select a DEV with a low payment cost within its reachable range and send a charging request message. The charging request message is transmitted by the recorded relay vehicle node of Algorithm 1. Further details are discussed in section 4.3.2. The charging request message includes the remaining available power SoC_i, the required power amount a_i, and path. Once the charging request message is sent, the CEV starts to wait for the reply message from the DEV. If the CEV does not receive the acknowledgment message within a time slot t_w, it is regarded as rejected.

Algorithm 1 : The discharging information transmission routing algorithm.

```
de-the DEV; R_j-the distance between the DEV and the travel destination; d-the distance between the DEV and the packet carrier vehicle; DI-discharging information message;
FR={(the sequence number, relay vehicle ID)}-The forwarding path
Initialize FR={(0,de)}, CZ=00
while d ≤ R_j do
    Forward DI to one-hop vehicles
    if vehicle receives DI then
        Update CZ=11
    end if
    if the packet carrier vehicle is de then
        if de is on road segment then
            if DC=11 then
                Calculate S based on Eq. (1)
                Select relay vehicle based on Case 1.1
            else if DC=10 then
                Select relay vehicle based on Case 2.1
            else if DC=00 then
                Carry and forward
            end if
        else if de is at intersection then
            Select next-hop relay vehicle based on Case 4
        end if
    else if the packet carrier vehicle is relay vehicle then
        if relay vehicle is on road segment then
            if DC=11&CZ=00 then
                if CA=10 then
                    Select next-hop relay based on Case 1.2(a)
                else if CA=00 then
                    Select next-hop relay based on Case 1.2(b)
                end if
            else if DC=10&CZ=00 then
                if CA=11 then
                    Select next-hop relay based on Case 2.2(a)
                else if CA=00 then
                    Select next-hop relay based on Case 2.2(b)
                end if
            else if DC=00 then
                Carry and forward
            end if
        else if relay vehicle is at intersection then
            if CZ=00 then
                Select next-hop relay based on Case 5
            end if
        end if
        if next-hop relay vehicle is selected then
            Add (sequence number+1, its ID) to FR
        end if
    end if
end while
```

4.3.1. Selection CEV

After receiving the charging request message, the EVP_j should determine a EVC_i for V2V energy sharing. EVP_j keeps track of all received request messages for a duration τ_r from the time the discharging information message is sent. The duration τ_r is calculated as follows,

$$\tau_r = 2\frac{R_j}{r}t_{top}, \tag{3}$$

where t_{top} is the average wireless transmission delay per hop. If EVP_j receives only one request message, it evaluates its available power amount to determine whether it can fulfill EVC_i's charging request. If $b_j \geq a_i$, EVP_j will accept the charging request message and respond with a charging acknowledgment message. Otherwise, EVP_j may decline the charging request. If the EVP_j receives multiple request messages, which is the case when multiple CEVs choose the EVP_j, the EVP_j selects a CEV considering the individual utility and the SoC level of the CEV.

The travel time of the EVC_i to the road segment (m,n) can be expressed as,

$$T(p^i) = \sum_{(m,n)\in p^i} t_{mn}, \tag{4}$$

here, $T(\cdot)$ is a function to calculate travel time. p^i is the path of the EVC_i from the location when the charging request message is sent to the road segment (m,n). $(m,n) \in P^j$, P^j represent the travel path of the EVP_j to the specific destination.

The travel time on the road section is affected by the traffic flow density of the road section. Higher traffic flow density leads to longer travel times. The travel time is expressed as follows [29],

$$t_{mn} = [1 + \theta_1(\frac{q_{mn}}{q_{mn}^{max}})^{\theta_2}]\frac{L_{mn}}{v_{mn}}, \tag{5}$$

where q_{mn} and q_{mn}^{max} are the number of current vehicles and maximum vehicles on the road segment (m,n), respectively. $\frac{q_{mn}}{q_{mn}^{max}}$ is the traffic flow density of the road segment (m,n). v_{mn} are the free travel speed on the road segment (m,n). L_{mn} is the length of the road segment. θ_1 and θ_2 are adjustment parameters. When there are traffic jams, accidents, and road closures on a segment (m,n), the travel time of this segment will become very large.

The time difference between EVC_i and EVP_j travel to the road segment (m,n), which can be expressed as,

$$\mathcal{T}_{i,j} = |T(p^i) - T(p^j)|, \tag{6}$$

where $T(p^j)$ is the travel time to the road segment (m,n) of the EVP_j. To reduce waiting time and the impact on traffic behavior, the time difference $\mathcal{T}_{i,j}$ should be as small as possible.

The distance that EVC_i's current SoC can travel is greater than its travel distance to (m,n),

$$\frac{(SoC_i - TC)\times B_i^c}{\lambda^i} \geq Dis(p^i), \tag{7}$$

where SoC_i is the current SoC of EVC_i, TC is the minimum charge threshold. B_i^c is the battery capacity (in kWh). λ^i is the energy consumption rate per kilometer of the EVC_i. $Dis(\cdot)$ is a function to calculate travel distance. The corresponding travel distance $Dis(p^i)$ can be expressed as,

$$Dis(p^i) = \sum_{(m,n)\in p^i} L_{mn}. \tag{8}$$

The power amount b_j that can be sold by the EVP_j is greater than or equal to the power amount a_i requested by the EVC_i,

$$b_j \geq a_i. \tag{9}$$

The DEV prefers a requester which offers the high discharging utility, that maximizes this individual utility. The EVP_j calculates the discharging utility from the received charging request message, which can be expressed as follows [18],

$$P_i^j = p_c^j a_i - p_o a_i/\eta - p_t(T(p^j) + \mathcal{T}_{i,j}) - \gamma^j\sigma^j a_i, \tag{10}$$

where p_c^j is the selling price per unit power set by EVP_j, p_o is the original price per unit power, p_t is the time value computed in dollars/hour. η represents the V2V power transfer efficiency. γ^j is the battery replacement cost of EVP_j. σ^j is the capacity degradation coefficient of EVP_j. Therefore, $\gamma^j\sigma^j a_i$ is the battery degradation cost of EVP_j [30].

Moreover, the EVP_j not only needs to consider the individual utility but also the remaining SoC of the CEV when selecting the appropriate requester. For example, for two requesters EVC_1 and EVC_2, if $P_1^j > P_2^j$ but $SoC_1 > SoC_2$, the EVP_j cannot discard EVC_2. This is because the EVP_j needs to maintain the V2V energy sharing market order while obtaining the discharging utility. By considering the remaining SoC level of the CEV, it's possible to prioritize service for EVs with lower

charge levels during emergency charging situations, thus enhancing the system's responsiveness to urgent needs. Therefore, we aim to ensure discharging utility while taking into account the urgency of CEV demand, which is defined as the weighted sum of individual utility ratio and 1-SoC of the CEV, as follows,

$$\boldsymbol{SN_i^j} = \lambda_1 \frac{P_i^j}{p_c^j B_i^c} + \lambda_2 (1 - SoC_i), \tag{11}$$

where $\frac{P_i^j}{p_c^j B_i^c}$ represent the individual utility ratio. λ_1 and λ_2 are weight values, where $0 \le \lambda_1 \le 1$ and $0 \le \lambda_2 \le 1$, $\lambda_1 + \lambda_2 = 1$. The weight values λ_1 and λ_2 enable DEVs to balance their preference for discharging utility and consideration of the SoC for CEVs. For instance, if the DEV prioritizes its own discharging utility, higher values can be set for λ_1 (e.g., $0.5 < \lambda_1 \le 1$) and lower values for λ_2 (e.g., $0 \le \lambda_2 < 0.5$). Conversely, if the DEV prioritizes serving the CEVs, especially those with low SoC, higher values can be set for λ_2 (e.g., $0.5 < \lambda_2 \le 1$) and lower values for λ_1 (e.g., $0 \le \lambda_1 < 0.5$). Based on the above discussion, the selection CEV problem can be expressed as an optimization model with an objective function:

$$\arg\max_{EVC_i} \boldsymbol{SN_i^j}; \tag{12}$$

where $EVC_i \subset V_c$. $V_c = \{EVC_1, EVC_2, \cdots EVC_c\}$ is the set of CEVs that EVP_j receive charging request messages.

Subject to:

1) $b_j \ge a_i$
2) $\frac{(SoC_i - TC) \times B_i^c}{\lambda^i} \ge Dis(p^i)$
3) $p^i = \{(q,g), \ldots, (m,n)\}$
4) $p^j = \{(x,z), \ldots, (m,n)\}$
5) $P^j = \{(x,z), \ldots, (m,n), \ldots, (f,d)\}$.

$p^i = \{(q,g), \ldots, (m,n)\}$ represent the charging path for EVC_i. $p^j = \{(x,z), \ldots, (m,n)\}$ represent the discharging path for EVP_j. $P^j = \{(x,z), \ldots, (m,n), \ldots, (f,d)\}$ represent the travel path to destination for EVP_j. The optimization model aims to determine the most suitable CEV and the road segment to start the energy sharing while driving to maximize the V2V energy benefit $\boldsymbol{SN_i^j}$. The charging time required by the CEV can be expressed as,

$$T_i^c = \frac{a_i}{w}, \tag{13}$$

where w is the charging rate. The actual charging time for CEV i on road segment (m,n) is t_{mn}, as shown in Eq. (5). Therefore, CEVs can adjust their travel speed or increase the distance traveled alongside DEVs to meet their required charging time. We propose the selection CEV algorithm as Algorithm 2. The computational time of the selection CEV task performed on moving EVs is very limited and can be neglected for the purpose of simplifying the formulation model.

Algorithm 2 : Selection CEV algorithm.

```
CR-the charging request message
V_c-the set of CEV that EVP_j receive CR
τ_r-the duration receive CR
Initialize V_c = ∅
for τ_r do
    if received CR from EVC_i then
        Add EVC_i to V_c
    end if
end for
for EVC_i ∈ V_c do
    if b_j ≥ a_i then
        for (m,n) ∈ P^j do
            Calculate the Eq. (7)
            if (SoC_i − TC)×B_i^c / λ^i ≥ Dis(p^i) then
                Calculate SN_i^j according to Eq. (11)
                if the value is maximum then
                    Select the EVC_i and the (m,n)
                end if
            end if
        end for
    end if
end for
```

Algorithm 3 : The request and acknowledgment message transmission algorithm.

```
1: de-the DEV
2: ce-the CEV
3: FR-the forwarding path according to Algorithm 1
4: /* The message is transmitted from ce to de */
5: i = the maximum sequence number
6: while i = !0 do
7:     if the i-th vehicle can be communicate then
8:         Select the i-th vehicle from FR as relay vehicle
9:         i = i-1
10:    else
11:        Select the next-hop relay vehicle based on Eq. (1)
12:        Update relay vehicle ID
13:        Get the updated forwarding path FR
14:    end if
15: end while
16: /* The message is transmitted from de to ce */
17: i = 0
18: while i < the maximum sequence number do
19:    if the i-th vehicle can be communicate then
20:        Select the i-th vehicle from FR as relay vehicle
21:        i = i+1
22:    else
23:        Select the next-hop relay vehicle based on Eq. (1)
24:        Update relay vehicle ID
25:        Get the updated forwarding path FR
26:    end if
27: end while
```

4.3.2. Request and acknowledgment routing algorithm

In this section, we discuss the delivery of request and acknowledgment data packets for V2V energy sharing. To ensure connectivity of the communication link, we propose a request and acknowledgment message transmission algorithm, which is capable of dealing with link failures and keeping the network connected. As illustrated in Algorithm 3, data packets are transmitted between CEVs and DEVs following the vehicle nodes recorded in the forwarding path outlined in Algorithm 1. If the next-hop relay vehicle in the forwarding path falls outside the communication range of the packet carrier vehicle, it indicates a link failure by the relay vehicle. In such cases, the packet carrier vehicle re-selects the next-hop relay vehicle node from its one-hop neighbor vehicles based on the S value. In this way, Algorithm 3 manages link failures and maintains continuous network connectivity without the need for repeated re-selections. Lines 5 to 15 detail the process of delivering request and acknowledgment message data packets from CEVs to DEVs, while lines 17 to 27 outline the process of delivering acknowledgment message data packets from DEVs to CEVs.

5. Performance evaluations

In this section, we present the simulation environment and simulation results analysis. We simulate the proposed discharging driven energy sharing protocol (DDES) to evaluate the network performance and energy sharing performance.

5.1. Simulation environment setup

The simulations for the proposed protocol are employed in NS2 [31]. The use of SUMO to generate a simulation scenario for vehicular com-

Table 3
Simulation parameters.

Simulation Parameters	Value
scenarion length (m)	2000
scenarion width (m)	2000
speed of EVs (km/h)	30-60
number of EVs nodes	100-350
DEVs percentage	10%-40%
CEVs percentage	10%-40%
required power amount(KWh)	20-50
selling energy unit price (cent/kWh)	10-20
original energy unit price (cent/kWh)	8
time value computed ($/h)	10
charging efficiency coefficient	95%
communication range (m)	250
MAC protocol	802.11
CBR packet generation rate	0.1
$\alpha_1, \alpha_2, \alpha_3, \alpha_4$	0.4,0.1,0.1,0.4

munication, including road definitions, intersections, and vehicle behaviors, is detailed. This scenario is then exported as an NS2-compatible network scenario tcl file [32]. NS2 generates a trace file to record packet transmission details, while the nam file allows observation of node movement. Using AWK, data delivery delay and packet delivery ratio can be analyzed based on the records in the trace file [7]. The simulation area comprises a 2000 m × 2000 m grid, with adjacent intersection nodes spaced 500 m apart. Each road segment is two-way, with two lanes in each direction. Vehicles have a communication range of 250 m [25]. Reflecting an urban scenario, vehicle speeds in the simulation range randomly from 30 to 60 km/h. The energy unit selling price varies between 10 and 20 cents/kWh, while the original energy unit price remains fixed at 8 cents/kWh. Time value is calculated at 10 $/h [20], [21]. Initial SoC_i is assumed to be between 10% to 30%. A random allocation assigns varying percentages of DEVs and CEVs to evaluate the protocol performance, with CEVs requiring power amounts between 20 and 50 kWh. The simulations were conducted on a Windows 11 computer equipped with an AMD R7 processor and 16 GB of memory. Further simulation parameters are outlined in Table 3.

The evaluation of the DDES in simulation involves assessing both network performance and the performance of selected V2V energy sharing pairs. Network performance is measured using metrics such as end-to-end delay and packet delivery ratio. End-to-end delay refers to the time taken for a packet to be delivered from sender to receiver, while packet delivery ratio represents the percentage of successfully delivered packets out of the total sent. For the selected V2V energy sharing pairs, performance evaluation involves analyzing metrics such as average utility, the minimum SoC of the selected CEV, and allocation rate.

Utility: the average discharging utility computed by DEV in cents. Each DEV utility is computed in Section 4.3.1 as,

$$P_i^j = p_c^j a_i - p_o a_i/\eta - p_t(TR(p^j) + \mathcal{T}_{i,j}) - \gamma^j \sigma^j a_i. \tag{14}$$

The minimum SoC of selected CEV: The SoC of each EV is expressed as,

$$SoC = \frac{Q_{remain}}{Q_{rated}} \times 100\%, \tag{15}$$

where Q_{rated} represents the rated capacity of the battery (in kWh), and Q_{remain} represents the remaining capacity in the battery (in kWh).

Allocation rate: The allocation rate is expressed as,

$$AR = \frac{AN}{NP} \times 100\%, \tag{16}$$

where AN is the number of allocated V2V energy sharing pairs, and NP represents the maximum value between the number of DEVs and the number of CEVs.

For performance evaluation, we conduct a comparative analysis between the proposed energy sharing protocol and two existing methods: V2V charging allocation (V2VCA) [25] and V2V Distributed Charge Sharing (V2VDisCS) [27]. To ensure a fair comparison, we generate the discharging utility of V2V energy sharing pairs in all compared methods using the equations provided in Eq. (14). Additionally, we designate the meeting point as the road segment in the path of the DEV.

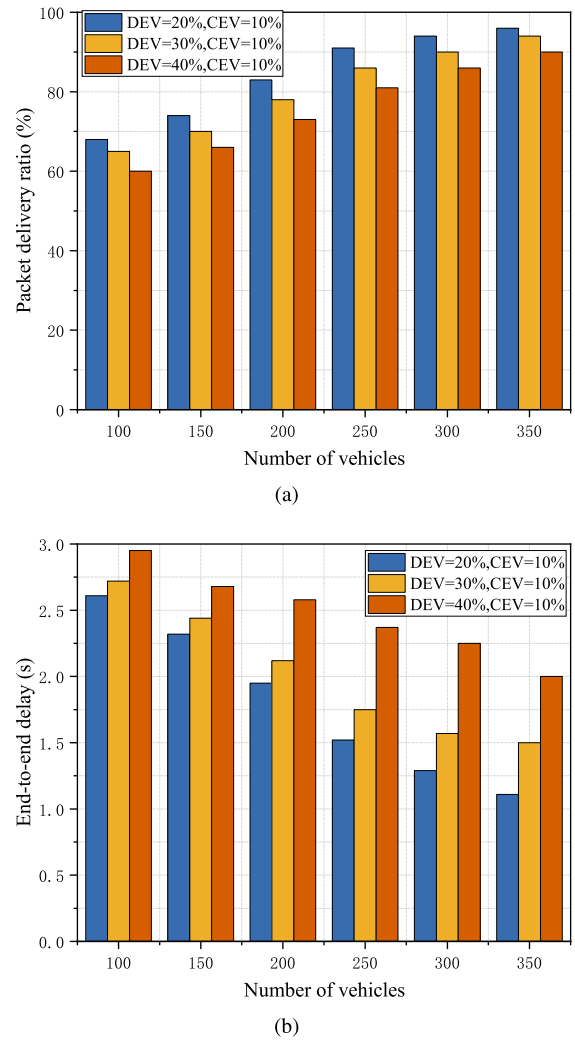


Fig. 10. Network performance results: (a) Packet delivery ratio vs. Number of vehicle. (b) End-to-end delay vs. Number of vehicle.

5.2. Evaluation of network performance

We define the maximum speed of vehicles as 60 km/h, while gradually increasing the number of vehicle nodes from 100 to 350. The percentage of CEVs to the total number of vehicles is set at 10%, and the percentage of DEVs to the total number of vehicles varies between 20%, 30%, and 40%.

Figs. 10(a) and 10(b) illustrate the packet delivery rate and end-to-end delay for three different percentages of DEVs across various total numbers of vehicles. As the number of vehicles increases, we observe a corresponding increase in the packet delivery rate and a decrease in end-to-end delay. This trend is due to the higher likelihood of vehicle connections with a larger vehicle count, providing more opportunities to select relay vehicles with improved performance and lower delay, consequently reducing packet loss rate and end-to-end delay. In Fig. 10(a), a decrease in the packet delivery rate is noticeable as the percentage of DEVs rises from 20% to 40%. This phenomenon occurs because as the number of DEVs increases, there's a corresponding increase in the number of packets that need to be delivered. However, the size of the vehicle buffer is limited, and with more packets to handle, there's a higher probability of data loss due to buffer overflow.

Fig. 10(b) illustrates that the end-to-end delay rises as the percentage of DEVs increases from 20% to 40%. This is because the increased number of packets leads to more queued transmissions, resulting in longer end-to-end delays.

We conducted simulations with 100 vehicles, varying their maximum speed from 35 km/h to 60 km/h. The percentage of CEVs to the total number of vehicles remained constant at 10%, while the percentage of DEVs varied as 20%, 30%, and 40% of the total number of vehicles. Figs. 11(a) and 11(b) depict the packet delivery ratio and end-to-end delay for the three different percentages of DEVs across various maximum speed scenarios. As the maximum speed of vehicles increases, we observe a decrease in the packet delivery rate and an increase in end-to-end delay for all three percentages of DEVs. This trend stems from the rapid changes in network topology induced by high vehicle mobility. Such changes lead to unstable communication links between vehicles and increased delays in packet delivery, ultimately reducing the packet delivery rate and extending end-to-end delays. In Fig. 11(a), we note a decline in the packet delivery rate with an increase in the percentage of DEVs. This decline is a consequence of the increased workload in transmitting data packets, resulting in a reduced packet delivery rate due to the constraints of the buffer size. Conversely, Fig. 11(b), illustrates that the end-to-end delay increases with a rise in the percentage of DEVs. This increase is attributed to longer queuing times for packet transmission as the number of packets grows, thereby extending the end-to-end delay.

From both Fig. 10 and Fig. 11, it's evident that under specific conditions, such as when there are 100 vehicles with a maximum speed of 35 km/h or when there are 350 vehicles with a maximum speed of 60 km/h, the packet delivery rate surpasses 95%. Furthermore, the end-to-end delay falls within the range of 1 to 2 seconds. These results indicate that the network can reliably transmit messages under certain vehicle speeds and numbers.

5.3. Evaluation of V2V energy sharing pairs performance

To evaluate the performance of V2V energy sharing pairs, we conducted simulations with 100 vehicles under two scenarios: (1) the percentage of DEVs fixed at 10% while the percentage of CEVs ranges from 10% to 40%, and (2) the percentage of CEVs fixed at 40% while varying the percentage of DEVs from 10% to 40%. To ensure a fair comparison, we employed two weights: $\lambda_1 = \lambda_2 = 0.5$ and $\lambda_1 = 1, \lambda_2 = 0$. The scenarios were structured to evaluate how changes in the percentages of CEVs and DEVs affect V2V energy sharing performance. In Fig. 12(a), we observe the discharging utility results as the percentage of CEVs increases from 10% to 40%, with DEV fixed at 10%. With the growing number of CEVs, the utility of all three schemes rises. This uptick is attributed to the increased availability of CEVs, providing DEVs with more options, particularly nearby CEVs with higher charging demands. While our scheme with weights set at $\lambda_1 = \lambda_2 = 0.5$ displays lower utility compared to V2V-CA schemes, it outperforms V2V-CA methods when the weights are adjusted to $\lambda_1 = 1, \lambda_2 = 0$. This suggests that our scheme can dynamically adjust the weights to accommodate varying V2V energy sharing demands based on DEV preferences.

In Fig. 12(b), we examine the discharging utility results as the percentage of DEVs increases from 10% to 40%, with CEVs fixed at 40%. With a higher number of DEVs, the utility of all three schemes diminishes. This decline can be attributed to the increased availability of DEVs, which provides CEVs with more options, including closer DEVs with lower energy prices. Consequently, the utility of DEVs is reduced. Moreover, it's worth noting that our scheme exhibits lower utility compared to V2V-CA when the weights are set to $\lambda_1 = \lambda_2 = 0.5$. However, with the weights adjusted to $\lambda_1 = 1, \lambda_2 = 0$, our scheme consistently achieves higher utility than V2V-CA. This observation underscores the scalability of our solution.

Fig. 13(a) illustrates the minimum SoC results of the selected CEVs when the percentage of DEVs is fixed at 10% and the percentage of CEVs increases from 10% to 40%. When $\lambda_1 = \lambda_2 = 0.5$, the minimum SoC level of the selected CEVs decreases as the number of CEVs increases. This decline is attributed to DEVs considering the SoC level of CEVs during selection. With more CEVs available, there are increased opportunities to select CEVs with lower SoC levels. Additionally, we observe that when $\lambda_1 = \lambda_2 = 0.5$, the minimum SoC level of the selected CEVs is lower compared to the other two methods. In Fig. 13(b), we present the minimum SoC results of the selected CEVs when the percentage of CEVs is fixed at 40% and the percentage of DEVs increases from 10% to 40%. Again, when $\lambda_1 = \lambda_2 = 0.5$, the minimum SoC level of the selected CEVs decreases as the number of DEVs increases. This trend reflects DEVs considering the SoC level of CEVs during selections. With more DEVs available, CEVs opt for energy from DEVs with lower energy prices, prompting DEVs to select CEVs with lower SoC levels to optimize overall utility. The minimum SoC level of CEV selected by the other two methods decreases because the number of CEVs served increases with the increase of DEV. When the number of CEV and DEV is 1:1, basically all CEV can be covered. Thus, the performance results of V2V energy sharing pairs indicate that considering the SoC level of CEVs during selection and adjusting different weights can lead to better individual utility or provide service preferentially for CEVs with lower SoC levels.

Fig. 14(a) illustrates the allocation rate results of V2V energy sharing pairs with 10% DEVs and varying percentages of CEVs between 10% to 40%. It is evident that the allocation rates of all four methods

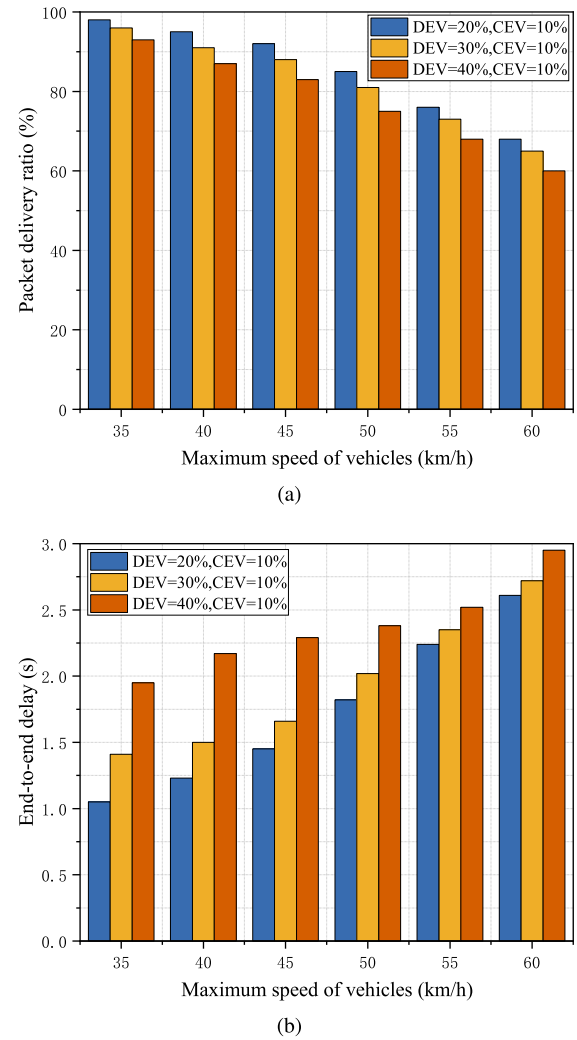


Fig. 11. Network performance results: (a) Packet delivery ratio vs. Maximum speed of vehicle. (b) End-to-end delay vs. Maximum speed of vehicle.

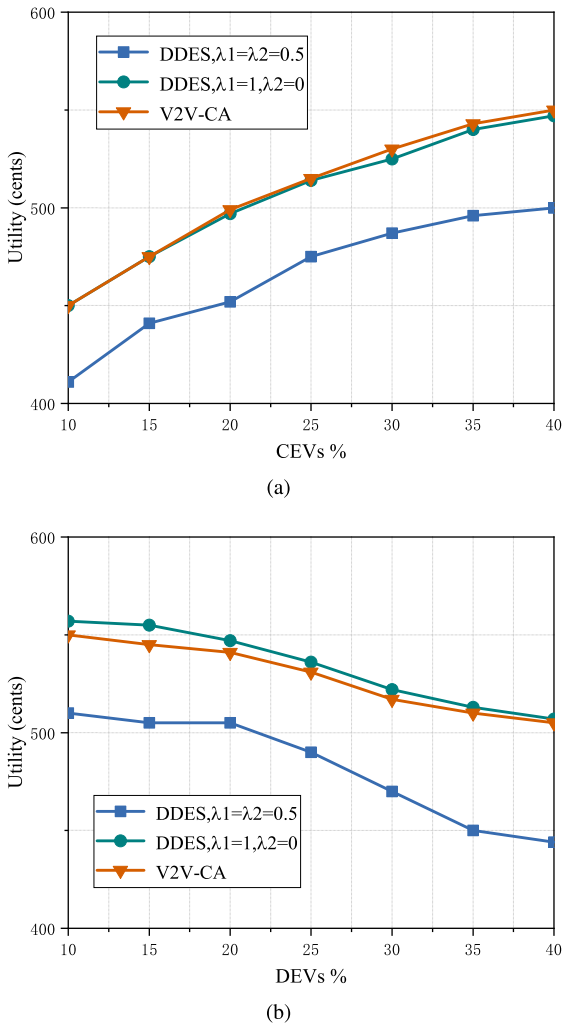


Fig. 12. Utility results: (a) Utility with 10% DEVs. (b) Utility with 40% CEVs.

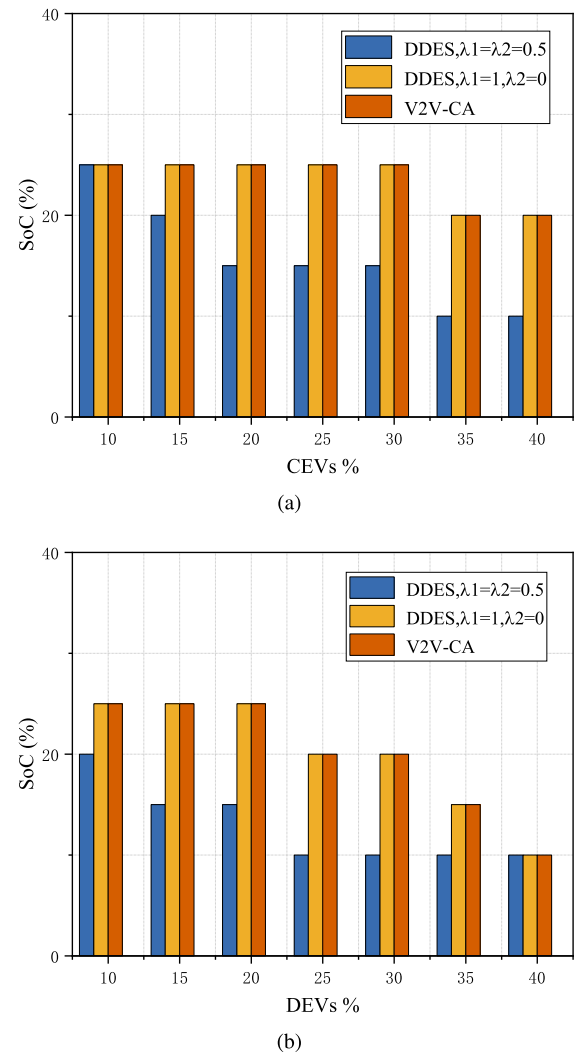


Fig. 13. Minimum SoC Level results: (a) Minimum SoC Level with 10% DEVs. (b) Minimum SoC Level with 40% CEVs.

increase with the percentage of CEVs. This phenomenon occurs because when the percentages of DEVs and CEVs are relatively low, the availability of energy from DEVs and the percentage of connectable DEVs limit the allocation of V2V pairs. As the percentage of CEVs increases, more opportunities arise for connectable DEVs to meet the demand, resulting in an increase in the allocation rate. Notably, the DDES method with $\lambda_1 = \lambda_2 = 0.5$ achieves higher allocation rates.

Fig. 14(b) presents the allocation rate results for V2V energy sharing pairs with 40% CEVs and DEV percentages ranging from 10% to 40%. It is observed that as the percentage of DEVs increases, the allocation rates of all four methods decrease. This occurs because when the percentage of CEVs exceeds that of DEVs, DEVs have more options, potentially achieving a 100% allocation rate. The allocation rates of the DDES method with $\lambda_1 = \lambda_2 = 0.5$ remain above 95%. Additionally, the allocation rates of the V2VDisCS method and the DDES method with $\lambda_1 = \lambda_2 = 0.5$ are on average 6.5% higher than those of the V2V-CA method and the DDES method with $\lambda_1 = 1, \lambda_2 = 0$. This suggests that our scheme can achieve higher allocation rates by adjusting the weights.

6. Conclusion

Incorporating V2V energy sharing into the vehicle charging system is vital for EVs, especially considering the critical role of SoC levels in EV mobility. In this paper, a discharging driven energy sharing protocol was proposed, where the DEV selects the CEV based on factors such as individual utility and CEV's SoC. This novel protocol takes into consideration the dissemination of discharging information, CEV selection, and acknowledgment processes. The dissemination of discharging information across various traffic scenarios, including intersections and areas between intersections, was discussed. Additionally, an optimization model for CEV selection was developed, enabling the DEV to choose the most suitable CEV and proceed with V2V energy sharing acknowledgment. The results demonstrate that the proposed protocol ensures reliable message delivery with a high packet delivery percentage and low end-to-end delay. Furthermore, the protocol outperforms benchmark schemes in serving CEVs with low SoC while maintaining individual utility and allocation rates. In future work, efforts will be made to enhance V2V energy sharing protocols by considering energy efficiency and sustainability in VEN, while addressing potential security attack issues. These advancements will improve the capabilities of the protocols and facilitate the evolution of VEN technology.

CRediT authorship contribution statement

Qinghua Tang: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Demin Li:** Conceptualization, Formal analysis, Funding acquisition, Supervision, Writing – review & editing. **Yihong Zhang:** Funding acquisition, Supervision, Writing – review & editing.

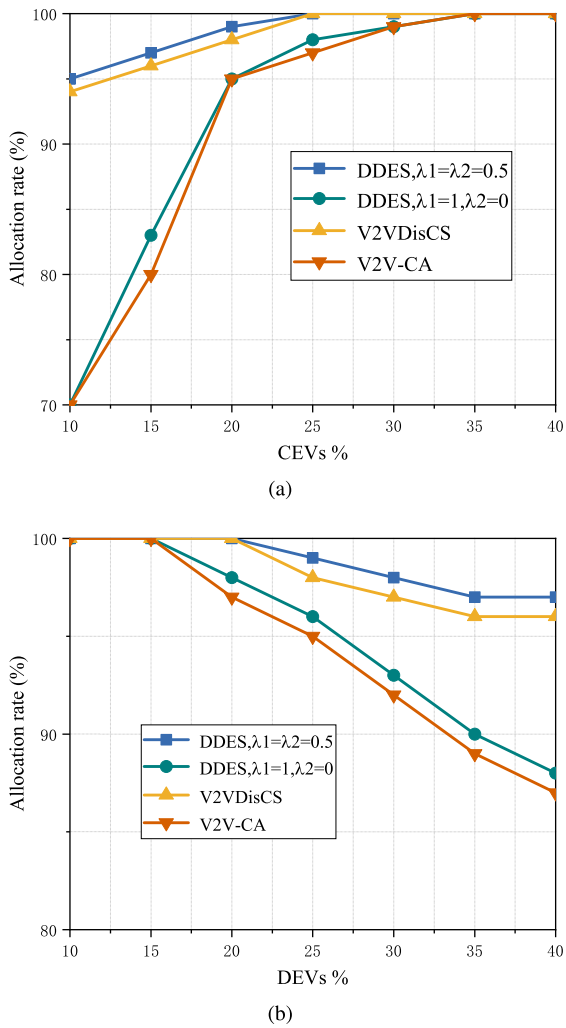


Fig. 14. Allocation rate results: (a) Allocation rate with 10% DEVs. (b) Allocation rate with 40% CEVs.

Xuemin Chen: Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

Acknowledgements

This work was supported by NSF of China under Grant No. 61772130, No. 71171045, and No. 61901104; the Innovation Program of Shanghai Municipal Education Commission, China under Grant No. 14YZ130; the International S & T Cooperation Program of Shanghai Science and Technology Commission, China under Grant No. 15220710600; the Fundamental Research Funds for the Central Universities and Graduate Student Innovation Fund of Donghua University CUSF-DH-D-2023040.

References

[1] A.Y. Lam, K.-C. Leung, V.O. Li, Vehicular energy network, IEEE Trans. Transp. Electrif. 3 (2017) 392–404.
[2] A.Y. Lam, V.O. Li, Opportunistic routing for vehicular energy network, IEEE Int. Things J. 5 (2017) 533–545.
[3] N. Mohamed, F. Aymen, T.E. Alharbi, C.Z. El-Bayeh, S. Lassaad, S.S. Ghoneim, U. Eicker, A comprehensive analysis of wireless charging systems for electric vehicles, IEEE Access 10 (2022) 43865–43881.
[4] L. Yang, M. Han, S. Fang, G. Wu, H. Sheng, H. Wei, X. Zhao, et al., Differentiated speed planning for connected and automated electric vehicles at signalized intersections considering dynamic wireless power transfer, J. Adv. Transp. 2022 (2022).
[5] Y. Wang, H.T. Luan, Z. Su, N. Zhang, A. Benslimane, A secure and efficient wireless charging scheme for electric vehicles in vehicular energy networks, IEEE Trans. Veh. Technol. 71 (2021) 1491–1508.
[6] A. Fitah, A. Badri, M. Moughit, A. Sahel, Performance of DSRC and WIFI for intelligent transport systems in VANET, Proc. Comput. Sci. 127 (2018) 360–368.
[7] C. Guo, D. Li, X. Chen, G. Zhang, An adaptive V2R communication strategy based on data delivery delay estimation in VANETs, Veh. Commun. 34 (2022) 100444.
[8] A.R. Khan, M.F. Jamlos, N. Osman, M.I. Ishak, F. Dzaharudin, Y.K. Yeow, K.A. Khairi, DSRC technology in vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) IoT system for intelligent transportation system (ITS), in: A Review, Recent Trends in Mechatronics Towards Industry 4.0: Selected Articles from iM3F 2020, Malaysia, 2022, pp. 97–106.
[9] L.A.L. da Costa, E.K. Duarte, M. Erneberg, E.P. de Freitas, A. Vinel, Poster: SafeSmart-A VANET system for efficient communication for emergency vehicles, in: 2020 IFIP Networking Conference (Networking), IEEE, 2020, pp. 643–645.
[10] A.M. Mezher, M.A. Igartua, Multimedia multimetric map-aware routing protocol to send video-reporting messages over VANETs in smart cities, IEEE Trans. Veh. Technol. 66 (2017) 10611–10625.
[11] P.B. Bautista, L.L. Cárdenas, L.U. Aguiar, M.A. Igartua, A traffic-aware electric vehicle charging management system for smart cities, Veh. Commun. 20 (2019) 100188.
[12] G. Li, Q. Sun, L. Boukhatem, J. Wu, J. Yang, Intelligent vehicle-to-vehicle charging navigation for mobile electric vehicles via VANET-based communication, IEEE Access 7 (2019) 170888–170906.
[13] Y. He, B. Venkatesh, L. Guan, Optimal scheduling for charging and discharging of electric vehicles, IEEE Trans. Smart Grid 3 (2012) 1095–1105.
[14] A.-M. Koufakis, E.S. Rigas, N. Bassiliades, S.D. Ramchurn, Offline and online electric vehicle charging scheduling with v2v energy transfer, IEEE Trans. Intell. Transp. Syst. 21 (2019) 2128–2138.
[15] M. Adil, M.P. Mahmud, A.Z. Kouzani, S. Khoo, Energy trading among electric vehicles based on Stackelberg approaches: a review, Sustain. Cities Soc. 75 (2021) 103199.
[16] Z. Teimoori, A. Yassine, A review on intelligent energy management systems for future electric vehicle transportation, Sustainability 14 (2022) 14100.
[17] M. Abdolmaleki, N. Masoud, Y. Yin, Vehicle-to-vehicle wireless power transfer: paving the way toward an electrified transportation system, Transp. Res., Part C, Emerg. Technol. 103 (2019) 261–280.
[18] R. Zhang, X. Cheng, L. Yang, Flexible energy management protocol for cooperative EV-to-EV charging, IEEE Trans. Intell. Transp. Syst. 20 (2018) 172–184.
[19] E. Bulut, M.C. Kisacikoglu, K. Akkaya, Spatio-temporal non-intrusive direct v2v charge sharing coordination, IEEE Trans. Veh. Technol. 68 (2019) 9385–9398.
[20] M. Shurrab, S. Singh, H. Otrok, R. Mizouni, V. Khadkikar, H. Zeineldin, An efficient vehicle-to-vehicle (v2v) energy sharing framework, IEEE Int. Things J. 9 (2021) 5315–5328.
[21] M. Shurrab, S. Singh, H. Otrok, R. Mizouni, V. Khadkikar, H. Zeineldin, A stable matching game for V2V energy sharing-a user satisfaction framework, IEEE Trans. Intell. Transp. Syst. (2021).
[22] H. Ko, T. Kim, D. Jung, S. Pack, Software-defined electric vehicle (ev)-to-ev charging framework with mobile aggregator, IEEE Syst. J. (2023).
[23] D. Said, H.T. Mouftah, Novel communication protocol for the EV charging/discharging service based on VANETs, IEEE Trans. Intell. Veh. 2 (2017) 25–37.
[24] D. Kosmanos, L.A. Maglaras, M. Mavrovouniotis, S. Moschoyiannis, A. Argyriou, A. Maglaras, H. Janicke, Route optimization of electric vehicles based on dynamic wireless charging, IEEE Access 6 (2018) 42551–42565.
[25] H. Abualola, H. Otrok, R. Mizouni, S. Singh, A V2V charging allocation protocol for electric vehicles in VANET, Veh. Commun. 33 (2022) 100427.
[26] F. Yucel, E. Bulut, K. Akkaya, Privacy preserving distributed stable matching of electric vehicles and charge suppliers, in: 2018 IEEE 88th Vehicular Technology Conference (VTC-Fall), IEEE, 2018, pp. 1–6.
[27] P. Chatterjee, P. Majumder, A. Debnath, S.K. Das, Distributed decision making for v2v charge sharing in intelligent transportation systems, in: 2022 19th Annual IEEE International Conference on Sensing, Communication, and Networking (SECON), IEEE, 2022, pp. 506–514.
[28] S. Zhou, D. Li, Q. Tang, Y. Fu, C. Guo, X. Chen, Multiple intersection selection routing protocol based on road section connectivity probability for urban VANETs, Comput. Commun. 177 (2021) 255–264.
[29] A. Skabardonis, R. Dowling, Improved speed-flow relationships for planning applications, Transp. Res. Rec. 1572 (1997) 18–23.

[30] S. Beer, T. Gómez, D. Dallinger, I. Momber, C. Marnay, M. Stadler, J. Lai, An economic analysis of used electric vehicle batteries integrated into commercial building microgrids, IEEE Trans. Smart Grid 3 (2012) 517–525.

[31] D. Soumaya, S. Dardouri, B. Ridha, Performance evaluation of routing protocols for vehicular AD-HOC networks using NS2/SUMO, in: Web, Artificial Intelligence and Network Applications: Proceedings of the Workshops of the 34th International Conference on Advanced Information Networking and Applications (WAINA-2020), Springer, 2020, pp. 352–365.

[32] A. Kusari, P. Li, H. Yang, N. Punshi, M. Rasulis, S. Bogard, D.J. LeBlanc, Enhancing SUMO simulator for simulation based testing and validation of autonomous vehicles, in: 2022 IEEE Intelligent Vehicles Symposium (IV), IEEE, 2022, pp. 829–835.